

Recovery of proteasome activity in cells pulse-treated with proteasome inhibitors is independent of DDI2

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

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Reviewed preprint posted

October 27, 2023 (this version)

Sent for peer review

August 21, 2023

Posted to bioRxiv

August 4, 2023

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Abstract

Rapid recovery of proteasome activity may contribute to intrinsic and acquired resistance to FDA-approved proteasome inhibitors. Previous studies have demonstrated that the expression of proteasome genes in cells treated with sub-lethal concentrations of proteasome inhibitors is upregulated by the transcription factor Nrf1 (NFE2L1), which is activated by a novel DDI2 protease. Here we demonstrate that the recovery of proteasome activity is DDI2-independent and occurs before the upregulation of proteasome gene expression. The recovery requires protein translation, but the efficiency of translation of proteasomal mRNA does not increase upon proteasome inhibition. Based on this data, we propose that the increased efficiency of proteasome assembly is responsible for the recovery of proteasome activity.

eLife assessment

This study presents **important** findings on a transcription-independent component of the early recovery of proteasome activity from a short pulse of proteasome inhibitor treatment. While the evidence supporting this conclusion is **solid**, experimental support for the proposal of an alternative regulatory process operative at the level of proteasome assembly, is **incomplete**. Lacking is experimental quantification of the effect of proteasome inhibition on the rate at which newly synthesized subunits are assembled into proteasomes.

Introduction

The ubiquitin-proteasome pathway is the primary protein quality control pathway in every eukaryotic cell. By degrading numerous regulatory proteins, this pathway also plays a pivotal role in regulating many cellular functions such as cell cycle and gene expression. Malignant cells are more dependent on proteasome function than non-transformed cells because they divide rapidly and produce abnormal proteins at a higher rate than normal cells [1, 2]. Proteasome inhibitors (PIs), bortezomib (Btz), carfilzomib (Cfz), and ixazomib are approved for the treatment of multiple myeloma (MM) and mantle cell lymphoma (MCL). MM cells are exquisitely sensitive to

PIs because the production of immunoglobulins by these malignant plasma cells places an enormous load on the proteasome and other components of the protein quality control machinery [3–6].

In the clinic, Btz and Cfz are administered twice weekly as a bolus IV where they cause rapid inhibition of proteasome activity in the patients' blood but are metabolized rapidly [7]. Within an hour after the administration, PIs concentrations in blood drops below the levels needed to kill tumor cells *in vitro* [8]. Although Btz has a very slow off-rate and Cfz is an irreversible inhibitor, proteasome activity recovers within 24hrs [6, 8–10]. This activity recovery may explain discrepancies between robust activity against cell lines derived from various cancers, continuously treated with Btz [11], and lack of clinical efficacy except for MM and MCL. In addition, recovery of activity has recently been implicated in PI resistance in MM [12].

In cells treated with proteasome inhibitors, transcription factor Nrf1 (also known as TCF11, encoded by *NFE2L1* gene) upregulates transcription of genes encoding all proteasome subunits [13, 14]. When the proteasome is fully functional, Nrf1 is constitutively degraded in an ubiquitin-dependent manner [13]. When the proteasome is partially inhibited, the ubiquitylated Nrf1 is recognized by DDI2 (Damaged DNA-Inducible 1 Homolog 2), a novel ubiquitin-dependent protease that activates Nrf1 by a site-specific cleavage [15, 16]. Although knockdown of DDI2 blocks the PI-induced transcription of proteasome genes [15, 16], initial studies implicating DDI2 in the activation of Nrf1 did not determine whether DDI2/Nrf1-dependent transcription leads to the recovery of activity after clinically relevant pulse treatment with PIs. In this work, we asked whether DDI2 is involved in activity recovery after such treatment. Surprisingly, we found that the proteasome activity recovered in the absence of DDI2 when the transcription of proteasome genes was not upregulated. Activity recovery preceded the upregulation of proteasomal genes. This data demonstrates the existence of a novel, DDI2-independent pathway for the recovery of proteasome activity in PI-treated cells.

Results and Discussion

To analyze DDI2 involvement in the recovery of proteasome activity after treatment with PIs, we used commercially available clones of HAP1 cells, in which DDI2 was knocked out by CRISPR (Fig. 1a). We have analyzed three different clones that were generated by using two different gRNAs (Table S1). We treated cells for 1hr with a range of concentrations of Cfz and Btz, and then cultured them in drug-free media (Fig. 1b). We measured inhibition of proteasome $\beta 5$ site, which is the prime target of Cfz and Btz [2], immediately after the 1hr treatment, and 12hr or 24hr thereafter (Fig. 1c). In a parallel experiment, we determined cell viability 24hr after treatment using the CellTiter-Glo assay (Fig. 1c), which measures intracellular ATP levels. Initial inhibition of proteasome was observed at sub-lethal concentrations, and proteasome activity recovered in cells treated with such concentrations. Surprisingly, no differences in the recovery between wt and DDI2-KO clones were observed (Fig. 1c). Deletion of DDI2 did not affect recovery, despite inhibition of Btz-induced proteolytic activation of Nrf1 in DDI2-KO cells (Fig. 1d). Thus, while confirming that DDI2 carries proteolytic activation of Nrf1, these findings indicate that DDI2 is not involved in the recovery of proteasome activity in Btz and Cfz-treated HAP1 and DDI2 KO cells.

Next, we knocked down DDI2 by two different highly efficient siRNAs in two PI-sensitive triple-negative breast cancer cell lines, SUM149 and MDA-MB-231 (Fig. 1e). The knockdowns did not significantly affect the recovery of proteasome activity in cells treated with 100 nM Btz (Fig. 1f). Thus, the recovery of proteasome activity after pulse treatment with sub-toxic concentrations of inhibitors is DDI2-independent.

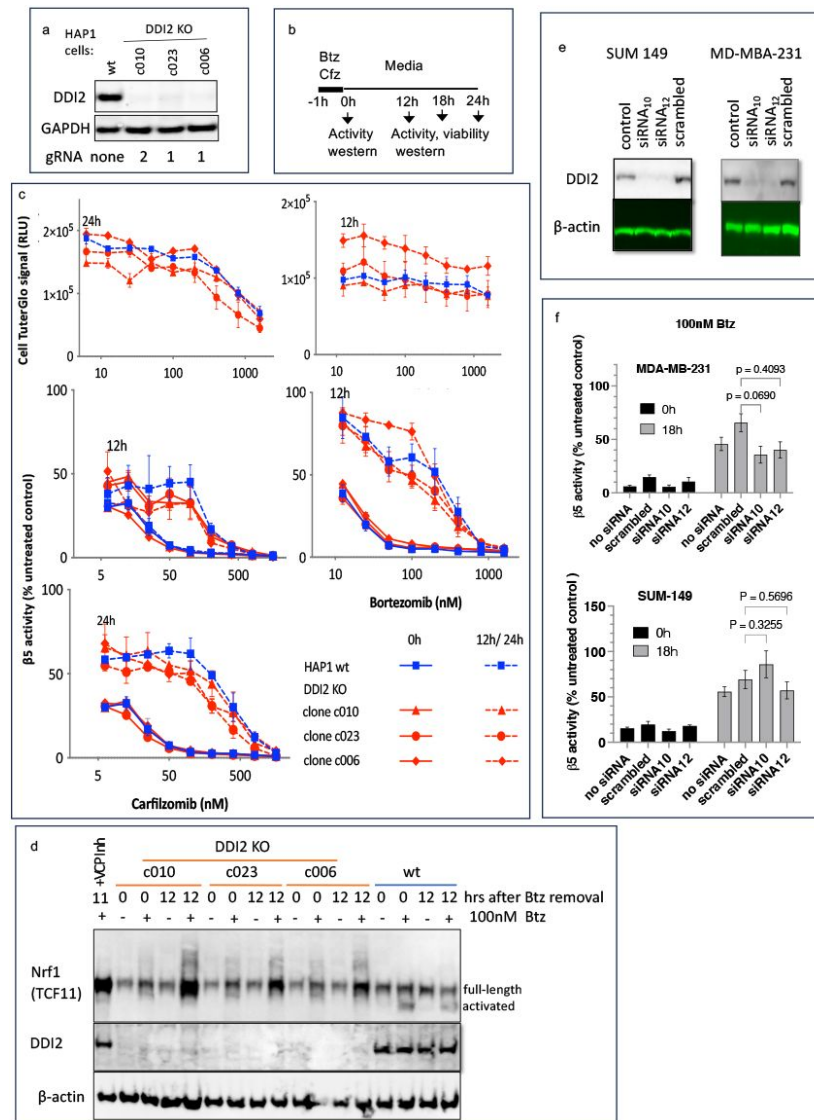


Figure 1.

Recovery of proteasome activity is DDI2 independent.

a. Expression of DDI2 in the CRISPR-generated clones of HAP1 cells used in this work was analyzed by western blot. The full-size membrane is presented in Fig. S1a. **b.** The experimental set up in this study. Cells were pulse treated with Btz and Cfz for 1hr, then cultured in a drug-free media, and analyzed at times indicated. **c.** Wt- and DDI2 KO clones of HAP1 cells were treated for 1hr. The cell viability was measured using CellTiter-Glo, and the inhibition of $\beta 5$ sites was measured with the Proteasome-Glo assay at times indicated. Data are averages $\pm$ S.E.M. of two to five biological replicates **d.** Knockout of DDI2 inhibits the Nrf1 processing. Western blots of Btz-treated HAP1 cells. Sample in the first lane are wt cells treated with VCP/p97 inhibitor CB-5083, which blocks Nrf1 processing [17–19], immediately after removal of Btz. The full-size membranes and proteasome activity measurements from this experiment are presented in Figs. S1b and S1c. **e.** MDA-MB-231 and SUM149 cells were analyzed by western blot 72hr after transfection with DDI2 siRNAs. The full-size membranes are presented in Fig. S2. **f.** The $\beta 5$ activity in siRNA-transfected SUM149 and MDA-MB-231 treated with 100 nM Btz for 1hr. The $\beta 5$ activity was measured using Suc-LLVY-AMC immediately and 18 hrs after treatment. Data are averages $\pm$ S.E.M. of three to five biological replicates. Statistical analysis was conducted using mixed-effect multiple comparisons; p-values ≤ 0.05 were considered significant.

If inhibitor-induced transcription of proteasome genes is responsible for the recovery of proteasome activity, the upregulation of proteasome gene expression should precede the activity recovery. However, we found that the recovery of activity started immediately after treatment and approached a plateau after 8hr (**Fig. 2a** [↗](#)), but first significant increase in the expression of proteasomal mRNAs occurred only 8hr after the removal of the inhibitor (**Fig. 2b** [↗](#)). These results suggest that the early recovery of the proteasome activity is not a transcriptional response.

Ruling out transcriptional response does not rule out the production of new proteasomes because protein synthesis can be regulated at the translational level. To determine whether the activity recovery involves biosynthesis of new proteasomes, we studied effects of the treatment with cycloheximide (CHX), an inhibitor of protein biosynthesis, on the recovery. Except for the first hour, the recovery was completely blocked by CHX, independent of DDI2 expression status (**Fig. 3a** [↗](#)). Thus, recovery of proteasome activity involves transcription-independent proteasome biogenesis.

Activation of mRNA translation could explain transcription-independent production of new proteasomes if a significant fraction of proteasomal mRNAs are untranscribed in the absence of PI treatment. We used polysome profiling to determine the distribution of proteasomal mRNA between translated and untranscribed fractions. We found that 90% of proteasomal mRNAs are ribosome or polysome bound in untreated cells (**Fig. 3b** [↗](#)), and that treatment with inhibitors did not increase this fraction (**Fig. S3**). This result agrees with a published result that the amount of proteasome mRNA in the polysomal fraction does not increase when proteasome is inhibited in MM1.S cells [20 [↗](#)]. Thus, the synthesis of new proteasomes in cells treated with sub-lethal concentrations of proteasome inhibitors occurs without upregulation of translation of mRNA encoding proteasome subunits.

How can the biogenesis of proteasomes be upregulated without an increase in the efficiency of mRNA translation? To gain activity, the catalytic subunits must assemble into the mature particles in a complex process involving multiple chaperones [21 [↗](#), 22 [↗](#)]. It is unknown what fractions of nascent subunits are incorporated into the mature proteasomes. One study found that proteasomes degrade a significant fraction of nascent proteasome subunits within 2 – 4hr following the synthesis [23 [↗](#)], after which the remaining fraction is highly stable (**Fig. 4a** [↗](#)). We hypothesize that nascent proteasome subunits are partitioned between immediate degradation and assembly, and the inhibition of the proteasome blocks degradation and increases the efficiency of proteasome assembly (**Fig. 4b** [↗](#)).

Increased proteasome assembly after treatment with proteasome inhibitors has been previously reported [24 [↗](#)]. If nascent polypeptides take ~1 – 2h to assemble into proteasomes, this model explains translation-independent recovery of proteasome activity in the first hour after removal of PIs (**Fig. 3a** [↗](#)). The fraction of nascent polypeptides that is degraded may be much larger than on **Fig. 4** [↗](#) because that experiment used 1hr pulse labeling, and was thus unable to detect nascent proteins that are degraded within minutes after synthesis. Thus, partitioning proteasome nascent polypeptides between degradation and assembly allows cells to instantaneously upregulate proteasome biogenesis immediately after proteasome inhibition.

The most important conclusion of this work is that, in addition to Nrf1/DDI2 pathway, mammalian cells possess at least one additional pathway to restore proteasome activity after treatment with PIs and that this DDI2-independent pathway is responsible for the rapid synthesis of new proteasomes immediately after treatment with PI. While this study was underway, two other laboratories found that knockout of DDI2 reduced recovery of proteasome activity in multiple myeloma and NIH-3T3 cells pulse-treated with PIs by ~30% [25 [↗](#), 26 [↗](#)]. Similarly, knockdown of Nrf1 did not completely block recovery of proteasome activity [14 [↗](#)]. These findings are consistent with the existence of the DDI2-independent pathway of recovery.

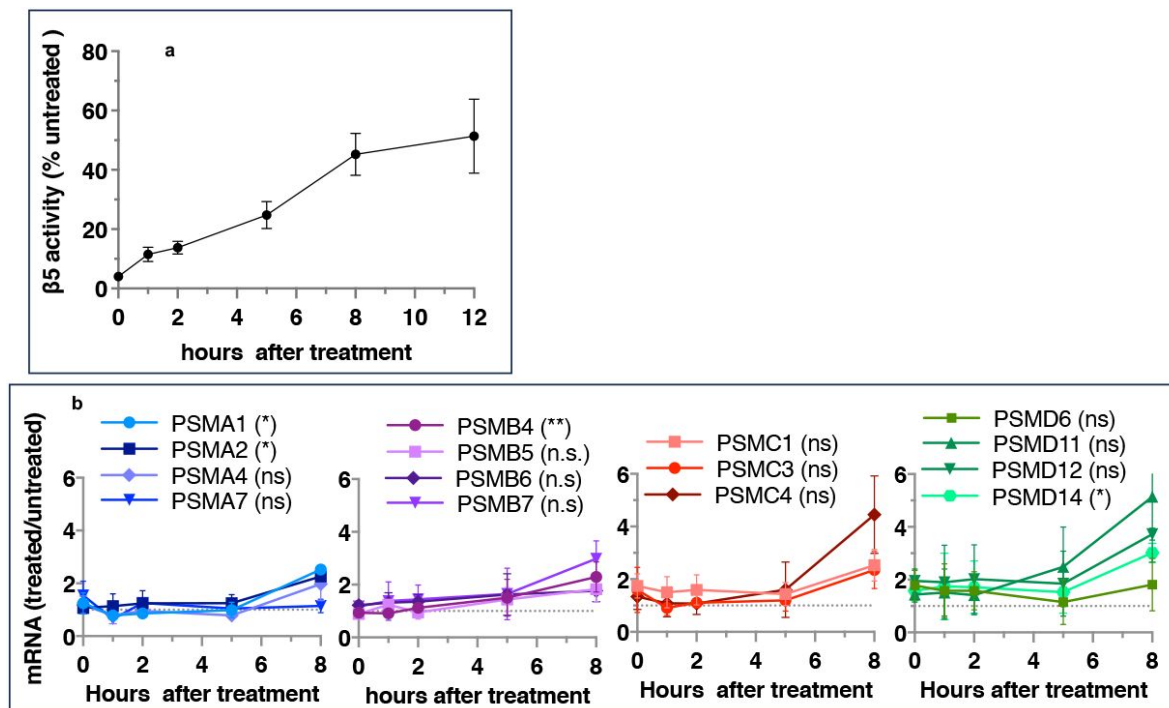


Figure 2.

Proteasome activity recovers before up-regulation of proteasome gene expression.

Wt-HAP1 cells were pulse-treated with Btz (100 nM), cultured in drug-free medium, and analyzed at indicated times. **a.** $\beta 5$ activity was measured using Proteasome-Glo and normalized first to CellTiter-Glo viability data, and then to proteasome activity in the mock-treated samples. The data are averages $\pm$ S.E.M. of two to five biological replicates. **b.** In a parallel experiment, the mRNA was isolated, and the expression of proteasome genes was quantified using quantitative RT-PCR. Data are averages $\pm$ S.E.M. of three biological replicates. The number in parenthesis indicates the t-test results at an 8hr time point.

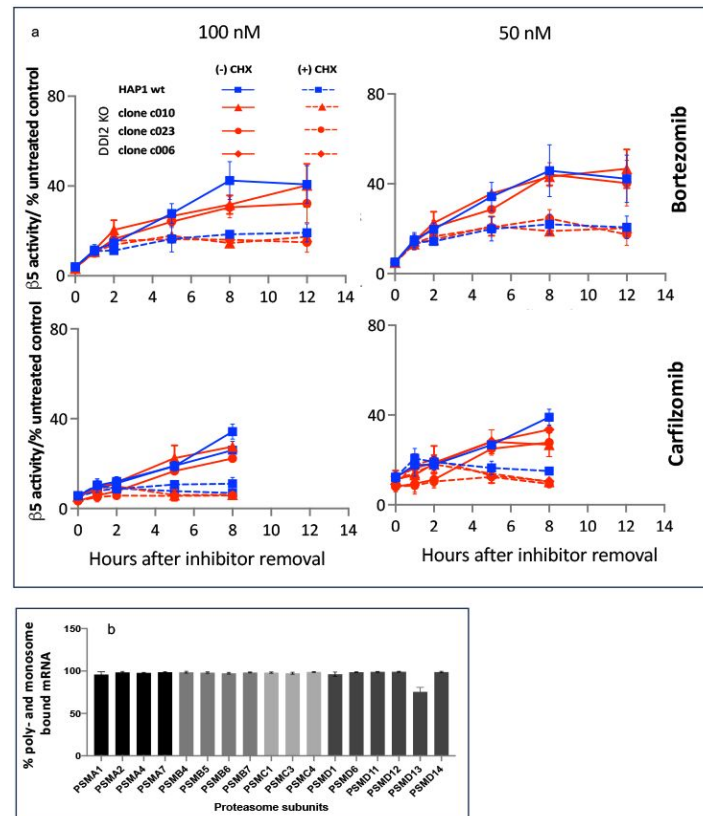


Figure 3.

The recovery of proteasome activity required the synthesis of new proteasomes.

a. Wt (blue) and DDI2 KO (red) HAP1 cells were treated for 1 hr at indicated concentrations of Btz and Cfz and then cultured in a drug-free media in the absence (solid line) or presence of CHX (dashed line). The $\beta 5$ activity was measured using Proteasome-Glo and normalized first to cell viability which was determined in a parallel experiment using CellTiter-Glo and then to untreated controls. Data are averages $\pm$ S.E.M of two to five biological replicates. **b.** All proteasome mRNAs are actively translated. mRNA isolated from untreated wt-HAP1 cells were analyzed by polysome profiling. The combined % mRNAs in the 80S and polysomal fraction relative to the cumulative amount of mRNA in all fractions are shown. Data are averages $\pm$ S.E.M of two biological replicates.

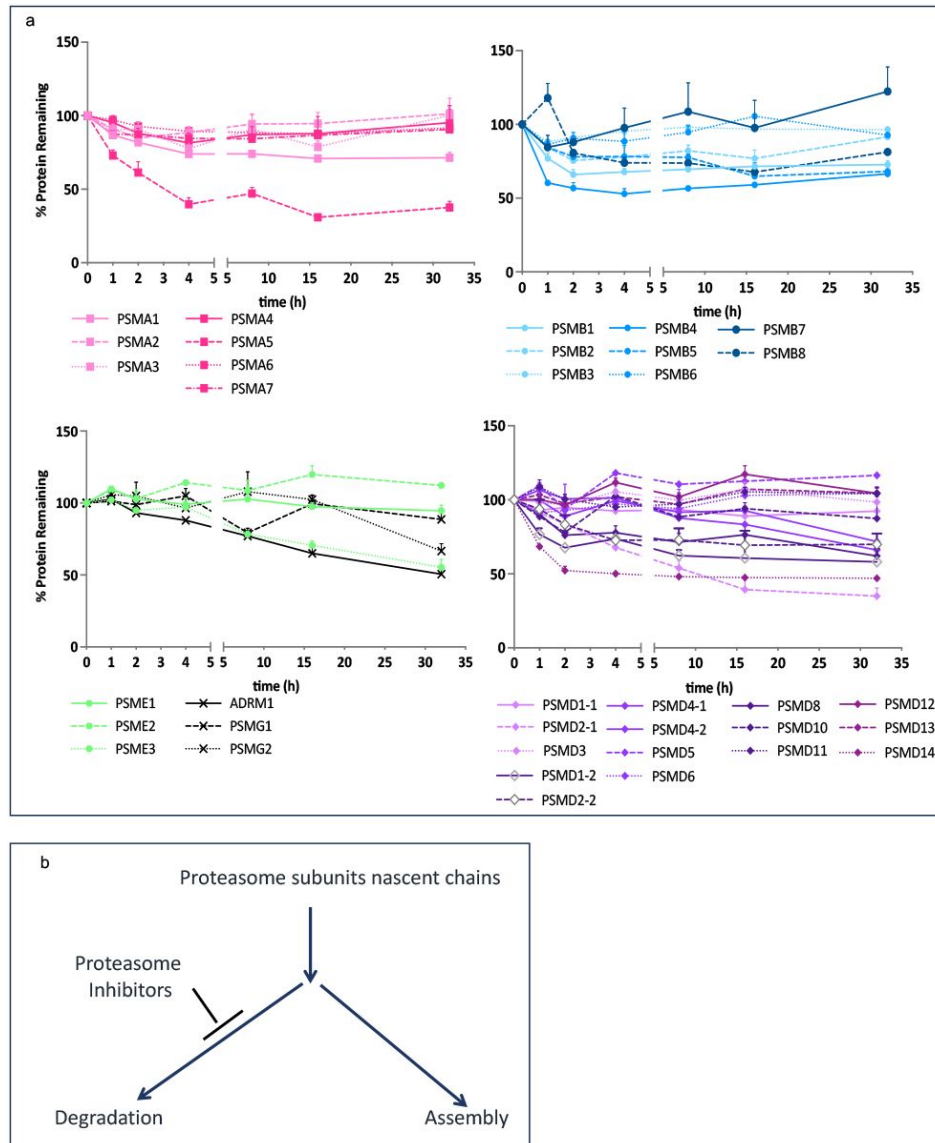


Figure 4.

Rapid degradation of nascent proteasome polypeptides can explain the rapid recovery of activity.

a. Turnover of proteasome subunit in human RPE cells was measured by quantitative mass-spectrometry following 1hr labeling with heavy isotopes. Data from Table S4 in [23]. **b.** Proposed model of how nascent proteasome subunits are partitioned between assembly and degradation.

These findings necessitate reconsidering the role of the DDI2/Nrf1 pathway in basal and inhibitor-induced proteasome expression. Previous studies have also reported that DDI2/Nrf1 contributes to the maintenance of basal levels of proteasomes [25, 27, 28], and that Nrf1 is essential for the basal proteasome expression in the brain [29], liver [30], and retina [31]; yet, in our experiments, the effect of DDI2 KO at the basal proteasome activity was not significant (Figs. 1f and S1c). These differences may reflect the fact that heavy secretory MM cells, embryonic cells, and certain specific tissues require higher levels of proteasome activity and use DDI2/Nrf1 pathway to supplement other pathways responsible for proteasome expression [32]. Other studies demonstrated importance of Nrf1-dependent proteasome expression during cardiac regeneration [33] and brown fat thermogenic adaptation [34].

Several groups have previously found that knockout of DDI2 sensitizes cells to proteasome inhibitors [10, 12, 25, 26, 35]. This was further interpreted as supportive of a role for DDI2-dependent recovery in de-sensitization of cells to PI-induced apoptosis. Although we confirmed this observation in HAP1 cells (not shown), the present finding suggests that DDI2 desensitizes cells to PI by a different mechanism. Activation of non-proteasomal Nrf1-dependent oxidative stress response genes [36, 37] may help overcome the deleterious consequences of PI-induced overproduction of reactive oxygen species (ROS) [38]. Alternatively, the ability of DDI2 to bind and participate in the degradation of large ubiquitin conjugates [35, 39] may help alleviate the stress associated with proteasome inhibition. DDI2 and proteasome are both involved in DNA repair [40–44] and impairment of proteasome in the absence of DDI2 can lead to excessive spontaneous DNA damage even in the absence of DNA-damaging agents. Finally, the proteolytic activation of another yet-to-be-identified DDI2 substrate cannot be ruled out. Thus, our study provides strong evidence for the existence of a novel pathway responsible for the recovery of proteasome activity in the inhibitor-treated cells, and, in combination with the published literature, should stimulate research on novel biological roles of DDI2, which can explain the embryonic lethality of DDI2 deletion [27] and DDI2 role in tumorigenesis [45].

Materials and Methods

Source of materials

HAP1 (wt-clone 631, DDI2 KO clones 006, 023, and 010, Table S1) were obtained from Horizon Discovery. MDA-MB-231 cells were purchased from ATCC® (Cat. #HTB-26™), SUM149 cells were obtained from Asterand [10]. Sources of inhibitors and other chemicals are listed in Table S2.

Cell culture

All cells were cultured at 37°C in a humidified atmosphere with 5% CO₂. HAP1 cells were cultured in Iscove's medium supplemented with 10% Fetal Bovine Serum (FBS). MDA-MB-231 and SUM149 cells were cultured in Dulbecco's Modified Eagle's Medium (DMEM)/Hams F-12 50/50 Mix supplemented with 5% FBS. SUM149 cell media were also supplemented with 4.8 µg/ml insulin, 10 mM HEPES, pH 7.3, and 1 µg/mL hydrocortisone. In addition, all media were supplemented with 100 µg/mL Penicillin-streptomycin, 0.2 µg/mL ciprofloxacin, and 0.25 µg/mL amphotericin B. Cells were plated overnight before treatment, then treated with inhibitors for 1hr in fresh medium. The inhibitor-containing medium was aspirated, and the cells were re-cultured in drug-free medium for times indicated when they were harvested and analyzed as described in figure captions. In siRNA knockdown experiments, the cells were transfected for 72hr before the treatments. The MDA-MB-231 or SUM149 cells were seeded in 6-well plates at 2×10⁵ cells/well the day prior to the transfections. The cells were transfected with DDI2 siRNAs (Table S3) at a final concentration of 25 nM using 0.3% DharmaFECT 1 (Cat. #T-2001-03, Dharmacon) in Gibco™ Opti-MEM 1X Reduced Serum Medium and Corning® DMEM/Ham's F-12 50/50 Mix without antibiotics and anti-mycotic.

Proteasome activity assays

The activity of proteasome $\beta 5$ sites was determined either by Suc-LLVY-AMC (7-amido-4-methylcoumarin) fluorogenic substrate or by the Proteasome-Glo™ assay (Promega), a luciferase coupled assay, which uses Suc-LLVY-aminoluciferin as a substrate [46, 47]. For the Proteasome-Glo assay experiments, the cells in 96-well plates were washed with PBS and lysed by one cycle of freezing and thawing in 25 μ L of cold PBS containing Digitonin 0.05%. 25 μ L Suc-LLVY-aminoluciferin and luciferase containing Proteasome-Glo reagent were added, thoroughly mixed, and preincubated on a shaker for ~10 minutes at room temperature (RT) before luminescence measuring using a mixture of PBS and Proteasome-Glo reagent as a blank. Each sample contained three technical replicates.

To determine the proteasome activity in the cell extracts, cells were lysed in ice-cold 50 mM Tris-HCl, pH 7.5, 25% sucrose, 0.05% digitonin, 2 mM EDTA, 100 μ M ATP, 1 mM DTT, and 1x PhosSTOP. The cells were incubated for 15 min on ice, centrifuged at $20,000 \times g$ for 15 – 20 min at +4°C, and the supernatants were used for the experiments. Protein concentrations were determined using Pierce™ Coomassie Plus (Bradford) Assay reagent (Cat. #23238) with bovine serum albumin as a standard. An aliquot of cell lysates containing 1 μ g of protein was spiked into a 100 μ L per well of the 26S assay buffer (50 mM Tris-HCl pH 7.5, 40 mM KCl, 2 mM EDTA, 1 mM DTT, and 100 μ M ATP) containing 100 μ M of Suc-LLVY-AMC. The mixture was thoroughly mixed and preincubated at 37°C for 10 minutes. An increase in fluorescence was monitored continuously at 37°C at the excitation wavelength of 380 nm and emission of 460 nm. The slopes of the reaction progress curves for three technical replicates were averaged, and the inhibition was calculated as a percentage by dividing the slopes of the inhibitor-treated samples with the slope of mock-treated controls.

Western blotting

Cells were lysed in an ice-cold lysis buffer containing 50 mM Tris-HCl pH 7.5, 0.5% CHAPS, 10% glycerol, 5 mM $MgCl_2$, 1 mM EDTA, 1 mM DTT, 100 μ M ATP, and 1x PhosSTOP™, and were centrifuged at $20,000 \times g$ for 15 – 20 min at +4°C. The protein concentration in the supernatants was determined using Pierce™ Coomassie Plus (Bradford) Assay reagent and used to normalize the sample loading. A mixture of samples with lithium dodecyl sulfate loading buffer was heated before fractionated on either NuPAGE™ Bis-Tris 8% Midi Gel (Invitrogen™, Cat. #WG1003BOX) or SurePAGE™ Bis-Tris 8% mini gel (GenScript, Cat. #M00662), using MES SDS running buffer (GenScript Cat. #M00677). The proteins were transferred on 0.2 μ pore-diameter Immobilon-pSQ PVDF membrane (Cat. #ISEQ00010) using Invitrogen™ Power Blotter 1-Step™ Transfer Buffer (Thermo Cat. #PB7300). The membrane was blocked with 5% Milk in TBST and probed with antibodies listed in Table S4.

RNA isolation and qPCR

The mRNA was isolated from cells using TRIzol™ Reagent (ThermoFisher Scientific Cat. #15596018) according to the manufacturer's protocol. Then, cDNA synthesis was performed using a High-Capacity cDNA Reverse Transcription kit (Applied Biosystems cat. #4368814). Before the qPCR run, the RNA and cDNA were quantified by UV absorbance using NanoDrop2000 (Thermo Scientific™). The Real-time qPCR was performed using 2x SYBR Green Bimake™ qPCR Master Mix on a Bio-Rad C1000 thermal cycler CFX96™ Real-Time System. The primers are listed in Table S4.

Polysome profiling was conducted according to a published procedure [48, 49]. Cells were washed in a cold PBS containing 100 μ g/mL CHX and were resuspended in the hypotonic buffer containing 5 mM Tris-HCl pH 7.5, 2.5 mM $MgCl_2$, 1.5 mM KCl, 1x complete mini protease inhibitor EDTA free, 100 μ g/mL CHX, 1 mM DTT, and 100 units RNasin Plus. Triton X-100 and Sodium deoxycholate were added to the cell suspension to a final concentration of 0.5%, followed by centrifugation at $20,000g$ for 15 – 20 min 4°C. Extracts were loaded on 5 – 55% gradients of sucrose in 20 mM HEPES-KOH, pH 7.6, 0.1 M KCl, 5 mM $MgCl_2$, 100 μ g/mL cycloheximide, 1x complete EDTA free protease inhibitor cocktail, and 100 units/mL RNasin®. Following the centrifugation at

35,000 rpm for 2.5 hours at 4°C, the gradients were manually fractionated into 200 µL fractions. Fractions containing the non-translated mRNA, 80S ribosomes, and polysomes were pooled. The RNA was isolated and quantified by quantitative RT-PCR.

Statistical analysis

All statistical analyses were carried out in GraphPad PRISM.

Acknowledgements

This work was supported by 5R01CA213223 grant from the NCI to AFK. Ibtisam was supported by the LPDP scholarship from Indonesia Endowment Fund for Education. The authors are grateful to Dr. Wade Harper, late Dr. Alfred Goldberg for advice, and to Tyler Jenkins for the critical reading of the manuscript.

Conflict of interest statement

AFK is a Co-founder and Chief Scientific Officer of InhiProt, LLC.

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Reviewer #1 (Public Review):

Summary:

There has been substantial prior work trying to understand the transcriptional control of proteasome expression as an adaptive response to proteasome inhibition. This field has been mired by fierce debates over the role of the protease Ddi2 in activating the transcription factor Nrf1/NFE2L1. As the authors of this manuscript point out, most of the previous research centers on the continuous treatment of cells with proteasome inhibitors rather than a brief pulse of inhibition that better models the situation when these drugs are used clinically. The authors find that the initial recovery of proteasome activity is independent of Ddi2 and involves a mechanism distinct from transcription. The authors intriguingly point to a model in which the assembly of proteasomes is regulated. If true, this would be a significant finding, but for now, this model remains more speculative.

Strengths:

The pulsed treatment of proteasome inhibitors is a strength of this lab that few others use. It better mimics the clinical use of these inhibitors and allows for a more detailed analysis of the initial response to inhibition. The authors have used multiple different clones of Ddi2 knockouts and siRNA against Ddi2 to rule out the necessity of Ddi2 in the early production of proteasomes when cells are inhibited with proteasomes. establishing a thorough knockout approach while also avoiding compensatory mutations. These experiments are well controlled showing both the levels of Ddi2 upon knockout or knockdown and demonstration that cleavage of Nrf1, one of two known targets of Ddi2, is impaired. However, it should be noted that even in the knockout residual bands for Ddi2 remain. Since these HAP1 cells only have one copy of the Ddi2 gene, it is possible that this other band could be Ddi1, a very similar paralogue. If so the conclusions of Ddi2-like activity with Ddi1 must be tempered and rely more on the data with Nrf1 knockdowns.

This article sensitively monitors the recovery of proteasome function with the $\beta 5$ activity assay and for the production of new proteasome transcripts by Q-PCR. This precision coupled with detailed analysis of the timing are strengths that pointed to a more rapid recovery than transcription alone.

Weaknesses:

This paper's major weakness is the difficulty in establishing the authors' model that assembly is regulating this process. They do a convincing job demonstrating that activity recovers

before transcription. The evidence that translation is not affected depends entirely on the polysome RNA profiling from two replicates. Clearer and orthogonal data would help establish this finding. The stability of subunits is interesting and important in its own right. However, the clustering of proteins is somewhat unusual. The authors include PSMB8, an immuno-proteasome subunit that is not regulated by Nrf1. The proteins highlighted in green are an unusual assortment of alternative activators (PSME1-3), a ubiquitin-binding protein (ADRM1), and proteasome chaperones (PSMG1-2). Similarly, the purple proteins are not just proteins in the 19S regulatory particle but also assembly chaperones. However, these labeling issues do not detract from the conclusions of this figure.

In short, the authors establish that Ddi2 is not necessary for the initial, non-transcriptional, recovery of proteasome activity after a pulse of proteasome inhibition.

It is not clear what clinical impact this work will have. Although it models the pulse of proteasome inhibition more perfectly, it only looks at a single pulse rather than multiple treatments. Thus, ruling Ddi2's importance out for clinical benefit may be premature. More significantly this work suggests that the assembly of proteasomes might be a regulated process worth substantial follow-up that will be interesting to follow.

Reviewer #2 (Public Review):

Summary:

In this work, Ibtisam and Kisselev explore the role of DDI2 in proteasome function recovery after a clinically relevant pulse dosing using different proteasome inhibitors and their corresponding PK properties. The authors report that despite the lack of NRF1 activation by DDI2 there was no difference in recovery from pulsed proteasome inhibition observed in DDI2 KO cells as compared to WT controls suggesting that DDI2 is not required for recovery in this system. They further show that transcription of the proteasome subunits is initiated only after partial recovery of proteasome activity is already observed suggesting that non-transcriptional mechanisms might be also involved. The authors further show that translation inhibition blocked the recovery from proteasome inhibitors.

Strengths:

Overall, it is very important and informative to use a pulse treatment type approach (mimicking the PK properties of the drugs) to explore the biology of PIs as used in this study. The authors also provide convincing data that DDI2 is not required for proteasome activity recovery post-PI pulse treatment in the systems they explored.

Weaknesses:

Many of the other conclusions are not supported by the data in the current form of the manuscript and are too speculative and ignore the major findings in the field that can present alternative mechanisms. In particular, the authors discuss the "levels" of the proteasomes post-PI treatment without measuring the actual protein level of the individual subunits or the different assembled proteasome complexes.

Reviewer #3 (Public Review):

Summary:

In their manuscript "Recovery of proteasome activity in cells pulse-treated with proteasome inhibitors is independent of DDI2", Ibtisam and Kisselev investigate proteasome recovery in HAP1 cells either WT or DDI2 KO upon inhibition of proteasome via bortezomib or carfilzomib. The authors argue that proteasome recovery is independent of DDI2 as it is independent of the novo proteasome subunit synthesis. They argue recovery is dependent on the assembly of already synthesized proteasome subunits.

Strengths:

The findings are important as they provide insight into a transcriptionally-independent proteasome stress recovery that is likely applicable across distinct cellular subtypes. Comparable proteasome recovery early on (<12 hours) from proteasomal inhibition in DDI2 KO cell lines was already noted in other manuscripts, including Chen et al, suggesting that this phenomenon is applicable to other histotypes.

Weaknesses:

Some of the conclusions are not adequately supported by the data and how generalizable these findings are is unclear. In particular, there is concern regarding the status of the ubiquitin-proteasome-system in the HAP1 cell line that was used for these studies. In a previously published model system, a dependency on DDI2 and NRF1 was clearly demonstrated and this pathway was critical for late (12-24 hours) proteasome recovery as well as cell viability. The model system used here (HAP1 cells) seems completely independent of DDI2 both for proteasome recovery and viability as curves are substantially overlapping. It would be important to assess how the baseline proteasome activity in HAP1 cells compare to other cell lines and model system as these cells may be largely independent of proteasome degradation and their synthetic load on the pathway very modest.

It would also be relevant to look at later time points of proteasome recovery as one would expect DDI2 to play a role later on in the recovery of proteasome. the authors may have missed that time point as cells do not appear to recover close to 100% proteasome activity by 24 hours not even when the smallest concentration of carfilzomib is used.

A critical experiment to look at de novo proteasome assembly was not carried out, leaving the data hypothetical.

Finally, the authors leverage HAP1 cells for their work and should be mindful of not generalizing findings or disputing other author's conclusions in the absence of adequate experiments to support their hypothesis.

Author Response

We are grateful to the reviewers for recognizing the importance of our work on transcription-independent early recovery of proteasome activity. We also thank them for their thoughtful criticisms and suggested improvements, which we will address in the revised version as described below.

The reviewers and editors asked for data to support the model that early recovery of proteasome activity is due to accelerated proteasome assembly. This model is backed by published data that proteasome assembly intermediates increase dramatically in cells treated with proteasome inhibitors (Fig. 6 in Ref. 46 of the revised manuscript). We will expand the discussion of this paper in a paragraph that describes our model. Another key experiment to confirm this model would be to determine what fraction of nascent polypeptides is degraded within minutes after synthesis, which is not trivial, and Ibtisam ran out of time to conduct these experiments because she had to graduate in spring before the expiration of her visa. This type of experiment usually uses metabolic labeling by a heavy or radioactive amino acid that always includes a prior depletion of a non-labeled amino acid. However, the fundamental flaw of this approach, which is not recognized by the scientific community, is that depletion of an amino acid stresses cells and reduces the rate of protein synthesis, especially if this amino acid is methionine. Thus, this model is not easily to test, and should be considered a speculation. We will therefore move the description of this model, together with Fig. 4, into a separate "Ideas and Speculation" section and remove this model's description from the abstract.

Reviewer 1 raised the possibility that a background band detected on the western blot of DDI2 KO cells could be a highly homologous protease DDI1. This is highly unlikely because, according to Protein Atlas, DDI1 is selectively expressed in the testis and is not expressed in the cell lines we used. Reviewer 1 also suggested that we should base our conclusion on Nrf1 KD, which we de-facto did because we confirmed that DDI2 KD blocks Nrf1 activation (Fig. 1d).

In response to Reviewer 1 critiques regarding the presentation of proteasome subunits stability data in Fig. 4 (Ref. 45 of the revised manuscript), we will remove PSMB8 and replace chaperons with the subunits of the 26S base. We will change color palettes, symbols, and axis scales to improve clarity.

We will acknowledge in the discussion that our work did not exclude DDI2 role in the recovery of proteasome after repeated pulse treatments, as suggested by Reviewer 1.

We agree with Reviewer 2 that using proteasome levels is inaccurate when describing our activity measurement data. However, in the manuscript, we use "levels" only when discussing data in the literature. We believe measuring activity and not the total levels is more important because not all proteasomes are active, e.g., latent 20S proteasome core particles.

Reviewer 3 expressed concern that our conclusions were based on data in HAP1 cells, which are haploid, and appear not very sensitive to proteasome inhibitors. This is why we used DDI2 KD in MDA-MB-231 and SUM149 cells, which are highly sensitive to proteasome inhibitors (Weyburne et al., Ref. 11). In our experience, full extent of proteasome inhibitor cytotoxicity is not revealed until 48hr after treatments, and viability determined at 12hr and 24hr as on Fig. 1c should not be used to determine sensitivity (it was used for activity assay normalization). We will add a new supplementary figure showing that HAP1 cells are as sensitive to proteasome inhibitors as MDA-MB-231 cells when cell viability is assayed 48hr after treatment (new Fig. S2). Another panel on this new figure will demonstrate that the baseline proteasome activity is very similar in HAP1, MD-MB-231 and SUM149 cells. We will also add data demonstrating that inactivation of DDI2 by mutation does not change the recovery of proteasome activity in HCT-116 cells (new Fig. 1g). Recovery in MDA-MB-231, SUM149, and HCT-116 cells was measured at 18hr, which is still within the 12 – 24hr window when other investigators observed partially DDI2-dependent recovery.

We have conducted an experiment in which we followed activity recovery for up to 72hr. We found that activity plateaued at 24hr and opted against the repeat because there were no changes. We feel that the manuscript should not include one biological replicate data. The fact that the recovery is incomplete and that cells seem to survive with lower levels of proteasome activity is interesting; however, investigating the molecular basis for this phenomenon is beyond the scope of the current project.

We were not disputing the conclusions of previous studies that DDI2/Nrf1 is responsible for enhanced expression of proteasomal mRNA in cells continuously treated with proteasome inhibitors. In fact, we confirmed that pulse-treatment causes similar increase (Fig. 2b). As for papers that measured activity recovery after pulse treatment, we objectively discuss our results in the context of these papers.

We will also respond to Reviewers' recommendations and minor points:

- We will review the revised version carefully to eliminate spelling and grammatical errors and typos.
- We will no longer refer to DDI2 as a novel protease, as suggested by Reviewer 1.

- We agree with Reviewer 2 that our CHX results do not necessarily mean that recovery involves translation of proteasomal mRNAs, and we will now conclude that proteasome recovery requires protein synthesis.
- We will revise Fig. 1c, 3a and 4a to improve clarity.
- We have stated in the caption that data in Fig. 4a comes from Table S4 in Ref. 45.
- We will accept an excellent suggestion of Reviewer 3 to change "recovery" to "early recovery" in the title.
- Regarding Reviewer 3 request to assay activity recovery at additional time points before 12hr, this was done in the cycloheximide experiment in Fig. 3A.
- Even if we assume that the differences in the observed recovery activity in MDA-MB-231 cells (Fig. 1f) are statistically significant, which may implicate DDI2 involvement in the activity recovery, the percentage is still small, suggesting that most activity recovery is DDI2-independent.
- We will tone down the statement "the present findings suggest that DDI2 desensitizes cells to PI by a different mechanism," replacing "suggest" with "raise a possibility."
- We will indicate that only Bortezomib is approved for mantle cell lymphoma.
- We will change the description of clinical dosing as suggested by Reviewer 3. We will add a reference on PK of subcutaneous bortezomib (Ref. 9), even though the review we quoted (Ref. 7) discussed subcutaneous dosing.